Beyond Tenant Isolation

A Flow-Level Testbed for Common-Mode Resilience in Cloud-Dependent Backhaul

Milan Nikolic, Independent Researcher (m_nikolic@yahoo.com)

14 November 2026

Abstract

Cloud outages rarely respect tenant boundaries. A provider control plane, identity service or AI endpoint fails once and every deployment behind it fails together. Current cloud isolation --- logical tenants, confidential virtual machines, regions and provider diversity --- cuts tenant-to-tenant, workload-to-host and region-to-region, but not the shared provider control plane. Multipath is often proposed as the complement, but paths that share a conduit or a control plane are one path with extra steps. We present a flow-level testbed that compares nine architectures under ten outage events with 30 paired seeds and 2,700 runs, with the hypotheses and thresholds frozen in version control before the simulator existed. The architectures range from a single provider to tenant isolation, SRLG-disjoint multipath, independent control planes, a one-way boundary with a tested fallback, an untested fallback and a global cooperative controller. Tenant isolation contains a tenant-shared gateway fault (100% vs 0%) but not an identity failure (0% vs 0%); three paths over one conduit fail together (0% vs 100% for SRLG-disjoint paths); a one-way boundary with a tested fallback holds 90% under an AI control-plane failure against 0% without it; the isolation-multipath interaction is +44.4 points; and a global controller loses 100 points when it fails with its primary path. A tested fallback is worth 34.5 points over an untested one on the same failure. The results support three claims: current cloud isolation is the wrong cut for common-mode failure; multipath must be independent, not merely multiple; and a controller is a dependency, not an abstraction above dependencies. Evidence is a stipulated flow-level scenario, not a production measurement.

Keywords: common-mode failure; cloud resilience; network isolation; multipath; data diode; tested fallback; SRLG; cooperative backhaul.

1. Introduction

In November 2026 two disruptions in the same week had different triggers and the same propagation signature. A national aviation ground stop followed a telecommunications outage [15,16]; a Microsoft cloud outage, reported as a failed network capacity upgrade, coincided with disruptions to AI services including Claude, ChatGPT and Grok [17,18]. One event sits in a legacy-adjacent telecommunications chain; the other in a modern cloud control plane. The contrast defeats the comfortable explanation that legacy technology is the problem. The modern stack reproduced the same category without the legacy: a shared node failed, and every consumer behind it failed together.

The structural cause is concentration. A dependency shared by many consumers, with no independent failure domain behind it, turns a local fault into a correlated outage. Let a service depend on a shared node d and an independent part i. Its availability is bounded by

$$A_{\text{service}} \;\leq\; A_d \cdot A_i.$$

Adding consumers to d increases scale, utilization and blast radius at the same time. It does not add an independent failure domain; it widens the one that already exists. This is why concentration is economically rational and structurally dangerous: the same decision that lowers unit cost raises correlated exposure.

Cloud architecture does have isolation, but it isolates the wrong boundary for this failure. Logical tenant isolation (VPCs, IAM, hypervisors) cuts tenant-to-tenant; confidential computing and TEEs cut workload-to-host [20]; regions and availability zones cut region-to-region; provider diversity cuts provider A from provider B [1-4]. None of these cuts tenant-to-provider, consumer-to-control-plane, or consumer-to-shared-dependency. The weak point was never tenant-to-tenant. The shared control plane was.

The engineering response has two halves, and they are not substitutes. **Multipath and diversity** keep service running while a path, link or provider degrades. **Isolation** stops a fault from propagating across a boundary at all. Multipath cannot cut a shared identity service; isolation cannot create capacity on the protected side. A tested fallback is the condition that makes either one real: an isolation boundary that leaves the protected side without capacity converts a provider outage into a local outage, and an unexercised fallback is a control arm, not a mitigation.

This paper turns that argument into a testbed and a comparison. We do not propose a new routing protocol. We ask a narrower, falsifiable question: under a common-mode event, which architectural choices actually preserve service, and how do they compare with the isolation categories current cloud architecture already provides?

Contributions.

1. A flow-level testbed with nine architectures, ten outage events, 30 paired seeds and 2,700 runs, with the hypotheses and thresholds frozen at commit `1c781f3` before the simulator existed.
2. A comparison between stylized current-cloud architectures --- single provider, multi-zone and multi-region paths, tenant isolation, independent control planes and a global orchestrator --- and the two-axis principles of isolation plus independent multipath plus a tested fallback.
3. Five pre-registered results: tenant isolation is the wrong cut for a shared control plane; multipath gain tracks path independence, not path count; a one-way boundary without local capacity is a partition; the isolation- multipath interaction is positive; and a global cooperative controller is a new common-mode carrier.
4. An open artifact: frozen protocol, simulator, benchmark, figures, raw CSV outputs and tests.

What we do not claim. This is not a measurement of AWS, Azure, Google Cloud or any production network. The current-cloud architectures in Table 1 are structural analogues, not reconstructions of a particular provider. The model is flow-level: it computes capacity and dependency conjunction, not packets, queues, TCP behaviour or hardware. Security is not evaluated. The two motivating outages are reported events; their root causes are not independently verified here, and no result depends on a particular root cause being correct.

2. Related Work

2.1 Failure domains and isolation

Cloud computing is defined by on-demand, pooled resources and rapid elasticity [1], and the major providers expose isolation through regions and availability zones [2-4]. Tenant isolation is the default boundary: a tenant's virtual network, identity policy and hypervisor namespace are separate from another tenant's. Confidential computing extends this to the host, placing a workload in a hardware-enforced trusted execution environment. These mechanisms are effective for the threat they address. They do not address a provider control-plane fault, because the control plane is outside the tenant boundary by construction.

Dependability taxonomy separates faults, errors and failures, and distinguishes independent from common-cause failures [5]. Cloud measurement studies document that many cloud incidents are configuration and software faults rather than hardware faults [6]. Chaos engineering formalises the practice of testing failure paths before they are needed [7]. Operational technology security guidance recommends segmentation at the IT/OT boundary [8], and data diodes provide hardware-enforced one-way transfer at that boundary [9]. Rail signalling provides a vendor-described existence proof of a data diode paired with commercial off-the-shelf servers [9]. We treat the data diode as an availability and attack-surface boundary, not as a cryptographic control.

2.2 Multipath and independence

Multipath transport is standardised in MPTCP [10], and equal-cost multipath routing is standard in datacentre fabrics [11]. Shared-risk link groups (SRLGs) are the conventional way to express that two links are not independent [12]. The multipath literature usually evaluates throughput, latency and failover; it less often asks whether the path set is independent of the *control* plane that selects it. A three-path set that reconverges on one identity service or one controller is a single failure domain with three next hops.

Cooperative backhaul adds a second form of concentration. Self-scaling pools capacity across links and tenants, which raises utilization, but a global cooperative controller is itself a shared dependency. The controller can become the common-mode carrier that the multipath was meant to remove. We make controller locality an explicit design variable rather than an implementation detail.

2.3 Testbeds and pre-registration

Packet-level testbeds are the right instrument for protocol behaviour. They are a poor instrument for a structural question that spans many architectures and outage types: the cost of building each architecture in a kernel is high, and the result is still tied to one kernel and one topology. Our earlier layering testbed followed the opposite route, using a model, a discrete-event simulation and a Linux namespace harness to separate structural from implementation effects [13]. The present study is deliberately flow-level: it isolates the dependency structure and the path-independence condition, and leaves packet-level behaviour to a later fidelity. Paper B in the same repository is a caution against assuming that a biologically inspired controller will beat a simpler structural mechanism [14]; the biological parallel in this paper is a hypothesis generator, not evidence.

2.4 Positioning

Prior work treats isolation and multipath as separate remedies: isolation is a tenancy or segmentation property, multipath is a transport or routing property. The claim tested here is that they are independent mitigations of the same failure, and that neither substitutes for the other. The comparison is against stylized current-cloud architectures, not against a new protocol. The contribution is the joint test, the frozen protocol and the measured interaction, not a claim of priority over the individual mechanisms.

3. Principles and Model

3.1 Six principles

The testbed encodes six principles. They are stated before the results because they determine what the model measures.

1. **Availability is a conjunction.** A tenant is served only if its data path, its control plane, its tenant boundary and, where required, its fallback are available at the same time.
2. **Multipath counts only across independent failure domains.** Paths that share a conduit, a provider control plane or a shared control service are one path with extra steps.
3. **Isolation trades capability for containment.** A cut that removes a shared dependency can also remove the capability that dependency provided. A boundary without local capacity is a partition, not a mitigation.
4. **A fallback is a tested state.** It has an activation probability, a delay and a capacity fraction. An untested fallback is a control arm.
5. **The controller is a dependency.** Self-scaling through a global cooperative controller concentrates the service it is meant to diversify.
6. **Common-mode events are sampled once per event, not per tenant.** A shared node fails for everyone at once; that is what makes it common-mode.

3.2 Availability model

A deployment is a dependency graph $G = (V, E)$. A path set P serves a tenant, with path capacity c_p, latency and an SRLG set. Two paths are independent when their SRLG sets are disjoint and their providers differ. The **path independence index** is

$$I(P) \;=\; \frac{\max\{|S| : S \subseteq P,\ S \text{ pairwise independent}\}}{|P|}.$$

Under independent path failures with availabilities a_p, the path-set availability is

$$A(P) \;=\; 1 - \prod_{p \in P} (1 - a_p),$$

but under a common-mode event the same expression is capped by the shared dependency's availability. The cap, not the product, is the object of interest here.

A tenant also requires its control plane. If the shared identity or DNS service is down, the path set is unusable even when every link is up; if the AI endpoint is on the availability path, an AI failure is a service failure. An isolation boundary is a cut C in G. When the cut removes a required dependency, the tenant serves at most its local fallback capacity c_{fb}:

$$A_{\text{iso}} \;=\; \min\left(1, \frac{c_{fb}}{D}\right),$$

where D is demand. A fallback with activation probability q and delay d contributes $q \cdot \min(1, c_{fb}/D)$ only after d; the delay is averaged into the event window. A global controller that fails reduces the usable path set to the primary path; a local controller does not.

3.3 Architectures

Table 1 lists the nine arms. The first five and A8 are stylized versions of current-cloud choices; A6 instantiates the proposed combination; A7 is a fallback-quality control; A9 is the isolation-only control.

Table 1. Architectures.

Arm	Paths	Tenant iso.	Control	One-way	Fallback	Controller
A1	1 shortest	no	shared	no	none	none
A2	3 shortest	no	shared	no	none	local
A3	3 disjoint	no	shared	no	none	local
A4	3 disjoint	yes	shared	no	none	local
A5	3 disjoint	yes	independent	no	none	local
A6	3 disjoint	yes	independent	yes	tested	local
A7	3 disjoint	yes	independent	yes	untested	local
A8	3 disjoint	yes	independent	no	none	global
A9	1 shortest	yes	independent	yes	tested	none

Shared control means the tenant requires global identity, DNS and the AI endpoint. **Independent control** means identity and DNS are local; the AI endpoint is still required unless the arm is one-way. **One-way** removes the AI endpoint from the availability path and serves path traffic at 0.9 efficiency (the stipulated cost of local policy). **Tested fallback** activates with probability 0.98 after 2 s at 0.50 Gbps; **untested** activates with probability 0.60 after 8 s at 0.30 Gbps. **Global controller** uses only the primary path when the controller is down; **local** reallocates across all surviving paths.

3.4 Events

Table 2. Outage events.

Event	Failure	What it tests
`none`	no failure	normal operation
`link`	`a1-p1` down	one path, independent failure
`srlg`	all `a1-p*` links down	shared conduit
`backhaul_cut`	all agg-to-provider links down	total path loss, fallback
`provider_cp`	`p1-cp` down	provider control plane
`idp`	global identity down	shared identity
`ai`	AI endpoint down	shared AI control plane
`tenant_gw`	shared tenant gateway down	tenant-shared fault
`controller_link`	controller and `a1-p1` down	controller locality
`ai_link`	AI endpoint and `a1-p1` down	combined control and path failure

All events start at $15 + U(0, 10)$ seconds and last $20 + U(0, 10)$ seconds in a 60 s run. The start and duration are drawn once per seed and shared by all arms.

3.5 Mapping to current cloud architecture

The mapping is structural, not a claim about a specific provider.

A1 is a single-region, single-provider deployment: one path, one control plane, no fallback. **A2** is a multi-zone or multi-region path set that shares a conduit and a global control plane: more next hops, one failure domain. **A3** adds SRLG-disjoint paths, which is the strongest path diversity current cloud offers, but keeps global identity, DNS and AI. **A4** adds tenant isolation, the default cloud boundary. **A5** adds independent identity and DNS, the strongest single-tenant control-plane design in the set, but the AI endpoint remains on the availability path. **A8** is the global orchestrator: a single cooperative controller that can reallocate across all paths and tenants.

A6 is the proposed architecture: disjoint paths, tenant isolation, independent control, a one-way boundary at the AI/control interface, a tested local fallback and local reallocation. **A9** isolates without multipath: the same boundary and fallback on a single path. A9 is the control that separates "isolation works" from "isolation is sufficient".

4. Testbed and Protocol

4.1 Scenario

Six tenants connect through three aggregation sites to three providers and two cloud regions. Each tenant has three edge-to-aggregation links (1.05 Gbps, 1.0-1.5 ms), each aggregation site has three provider links (1.20 Gbps, 4.0-7.0 ms), and provider-to-cloud links run at 1.50 Gbps (8.0-9.0 ms). All provider links from one aggregation site share an SRLG (agg-j), so one conduit failure takes all of that site's provider links. The path bottleneck is 1.05 Gbps; mean demand is 1.0 Gbps with a per-seed 5% jitter. A path is edge $\rightarrow$ aggregation $\rightarrow$ provider $\rightarrow$ cloud; its capacity is the minimum link capacity and its SRLG set is the union of link SRLGs.

Shared dependencies are global identity (`idp`), DNS (`dns`), the AI endpoint (`ai`), the tenant gateway (`mt-gw`) and the cooperative controller (`coop`). Each tenant has a local fallback. The scenario is a testbed for the structural hypotheses, not a model of a specific operator.

4.2 Metrics

For each arm, event and seed we record:

- **availability during the event**: mean served/offered over tenants and event ticks;

- **overall availability**: mean over the 60 s run;
- **recovery**: time from event start until a tenant first reaches 90% for three consecutive ticks, or the event duration if it never does;
- **blast radius**: fraction of tenants below 50% during the event;
- **path independence**: the index of Section 3.2;
- **fallback rate**: fraction of tenants whose fallback activated;
- **cost index**: 0.5 per path, plus 0.1 each for tenant isolation, independent control and one-way, plus 0.05 for a global controller.

The cost index is a stipulated accounting device, not a price. It makes the trade-off visible; it is not used to rank architectures.

4.3 Protocol and reproducibility

The hypotheses and thresholds are frozen in `common-mode-predictions.md` at commit `1c781f3`. The benchmark runs 30 seeds (base 2026), nine arms and ten events: 2,700 runs. Each seed draws one event, one demand vector and one fallback vector; every arm sees the same draws. The simulator is deterministic given a seed. The benchmark writes per-seed, summary and verdict CSVs under `output/common-mode/`. The full reproduction command is

```
py -3 tools/benchmark_common_mode.py --seeds 30
py -3 -m unittest tests.test_common_mode -v
py -3 tools/make_common_mode_figures.py
```

The 22 new tests pass, and the repository's full suite (433 tests, one skipped) passes at the time of writing. The figures in this paper are generated by `tools/make_common_mode_figures.py` from the summary CSV.

5. Results

5.1 Overall picture

Figure 1 shows availability for every arm and event. The pattern is not a single ranking. A1 is available only with no failure. A2 survives an independent link failure but not a conduit failure. A3 survives both path events but not identity or AI. A4 adds tenant isolation and survives the tenant gateway. A5 adds independent identity and survives identity, but not AI. A6 survives AI, link, conduit, provider control, identity and tenant faults, and serves 45.6% under a full backhaul cut. A8 behaves like A5 until its controller fails, then falls to 0%. A9 matches A6 on the common-mode events but is much weaker under an independent link failure.

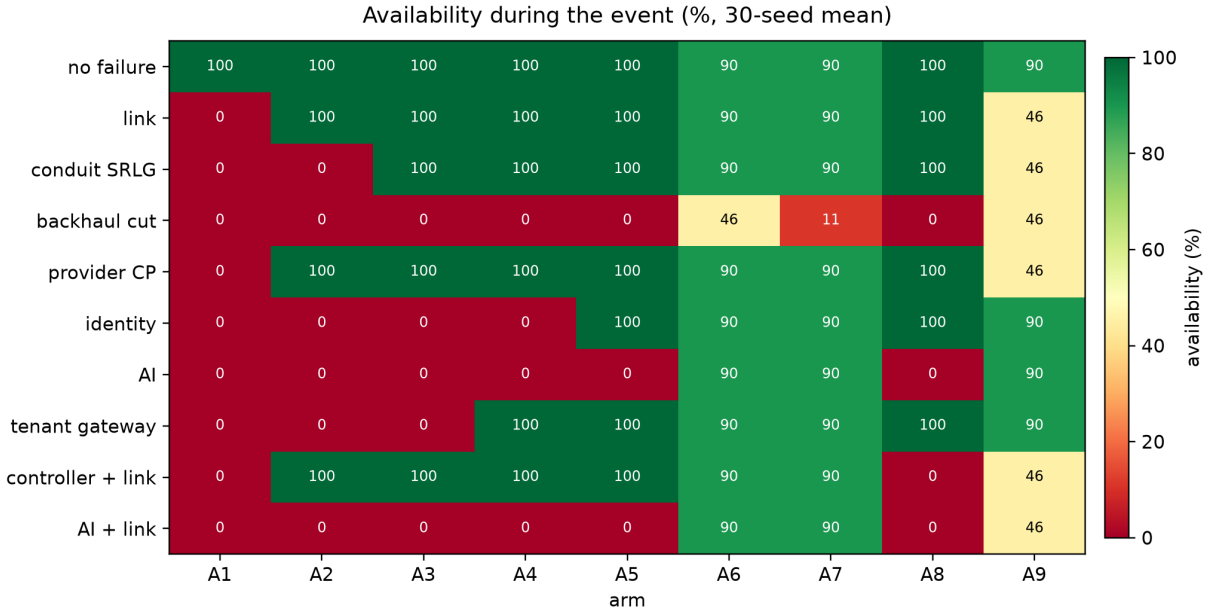


Figure 1. Availability during the event for the nine architectures and ten outage events (30-seed mean). Values are percentages; the color scale is shared. Data: `output/common-mode/common_mode_summary.csv`, regenerated by `tools/make_common_mode_figures.py`.

5.2 H16: tenant isolation is the wrong cut

Tenant isolation turns the tenant-shared gateway failure from a total outage into normal service: A3 is 0% and A4 is 100%. It does nothing for the identity failure: A3 and A4 are both 0%. The result is not that tenant isolation is useless; it is that the cut is in the right graph for tenant-to-tenant faults and in the wrong graph for the shared control plane. The shared node lies outside the tenant boundary, so no amount of tenant separation reaches it.

This is the category error in its cleanest form. Current cloud isolation is designed to prevent one tenant from reaching another; it is not designed to keep a tenant serving when the provider's identity or AI service is down. The fault that matters here crosses the boundary that tenant isolation does not cut.

5.3 H17: independence, not count

A2 and A3 both have three paths. Under the `agg-1` conduit failure A2 is 0%, because all three of its shortest paths share the conduit, while A3 is 100%, because its paths use three aggregation sites. Under the independent `a1-p1` link failure A2 is 100% and A1 is 0%. Path count alone would have predicted the opposite ordering for the conduit event.

The path independence index makes the difference explicit: A2 is 0.333 and A3 is 1.0. The same pattern appears for a provider control-plane failure: A1 is 0% and A2 is 100%, because A2 has paths via other providers. Multipath is a diversity property, not a multiplicity property. A path set that shares an SRLG or a control plane is a single failure domain with extra next hops.

5.4 H18: a boundary without capacity is a partition

Under the AI control-plane failure A5 is 0% and A6 is 90%. A5 has independent identity and DNS, but it still requires the AI endpoint; A6 removes that dependency with a one-way boundary and serves local policy at 0.9 efficiency. The normal-operation cost is 10 points: A5 is 100% and A6 is 90% with no failure. The cost is the price of local policy, not a failure of the boundary.

The fallback quality result is sharper. Under the full backhaul cut A6 serves 45.6% with a 99.4% activation rate; A7 serves 11.1% with a 56.1% activation rate. The tested fallback is worth 34.5 points on the same failure. An untested fallback is not a mitigation; it is a control arm that measures how often the organisation happened to configure the fallback correctly.

5.5 H19: the interaction is positive

Under the combined AI + link failure the four relevant arms are A1 = 0, A3 = 0, A9 = 45.6 and A6 = 90.0. The interaction term is

$$(A6 - A3) - (A9 - A1) = (90.0 - 0) - (45.6 - 0) = 44.38 \text{ points.}$$

The reading is direct. A3 has multipath but not isolation, so the AI failure takes it to 0. A9 has isolation but not multipath, so the link failure leaves it with fallback capacity only. A6 has both, and reaches 90. Isolation and multipath are not substitutes; under a mixed failure they are complements.

5.6 H20: the controller is a dependency

Under the controller + primary-link failure A5 (local reallocation) is 100% and A8 (global controller) is 0%. With no failure both are 100%, so the global controller buys nothing in normal operation and costs everything when it fails: once it is down, A8 falls back to its primary path, which is the failed link. This is the cooperative-controller trap. Self-scaling through a global controller raises utilization and recreates the concentration the multipath was meant to remove. Local, partition-tolerant reallocation avoids it.

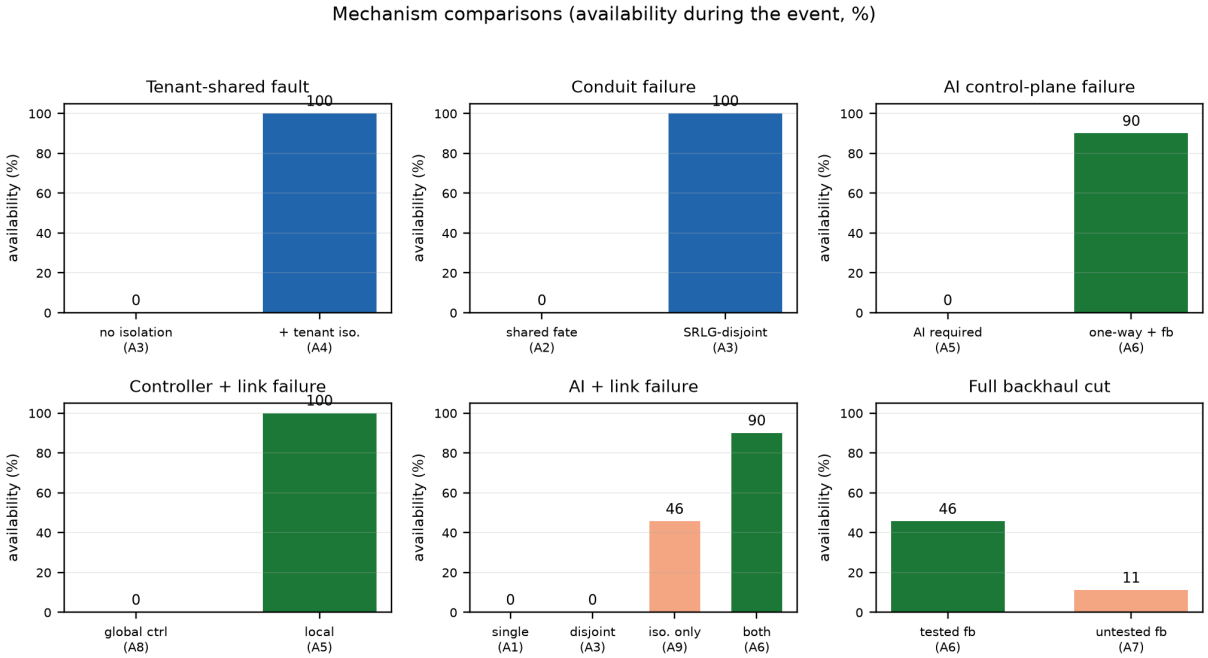


Figure 2. Mechanism comparisons. Each panel isolates one architectural choice: tenant isolation, path independence, the one-way boundary with fallback, controller locality, the combined interaction and fallback quality. Values are availability during the event (%); data as in Figure 1.

5.7 Comparison with current cloud architectures

Table 3 compares the stylized current-cloud arms with the proposed combination on the events that matter for common-mode failure. The table is the paper's central comparison. Arm identifiers follow Table 1: A1-A5 and

A8 are the current-cloud analogues, A6 is the proposed combination, A7 is the untested-fallback control and A9 is the isolation-only control.

Table 3. Current-cloud analogues versus the proposed combination (availability during the event, %).

Arm	Normal	Tenant gw.	Identity	AI	AI + link	Cut	Cost
A1	100.0	0.0	0.0	0.0	0.0	0.0	0.50
A2	100.0	0.0	0.0	0.0	0.0	0.0	1.50
A3	100.0	0.0	0.0	0.0	0.0	0.0	1.50
A4	100.0	100.0	0.0	0.0	0.0	0.0	1.60
A5	100.0	100.0	100.0	0.0	0.0	0.0	1.70
A6	90.0	90.0	90.0	90.0	90.0	45.6	1.80
A7	90.0	90.0	90.0	90.0	90.0	11.1	1.80
A8	100.0	100.0	100.0	0.0	0.0	0.0	1.75
A9	90.0	90.0	90.0	90.0	45.6	45.6	0.80

Current cloud improves along one axis at a time. Tenant isolation (A4) fixes the tenant gateway but not identity or AI. Disjoint multipath (A3) fixes the conduit failure but not the control plane. Independent control (A5) fixes identity but not AI. The global controller (A8) does not fix any common-mode event on its own, and it adds a new one. No current-cloud arm reaches 90% under AI + link; A6 does.

The comparison also exposes a trade that a single ranking would hide. A9 matches A6 on the four common-mode events and costs less (0.80 vs 1.80), because both use the same one-way boundary and tested fallback. A6's extra cost buys resilience to the independent link failure (90% vs 45.6%). If the operator's failure model is common-mode only, isolation plus a tested fallback is a defensible architecture; if independent path failures are in scope, multipath is still required. The "both axes" claim is therefore conditional on the failure mix, not a claim that the most expensive arm dominates everywhere.

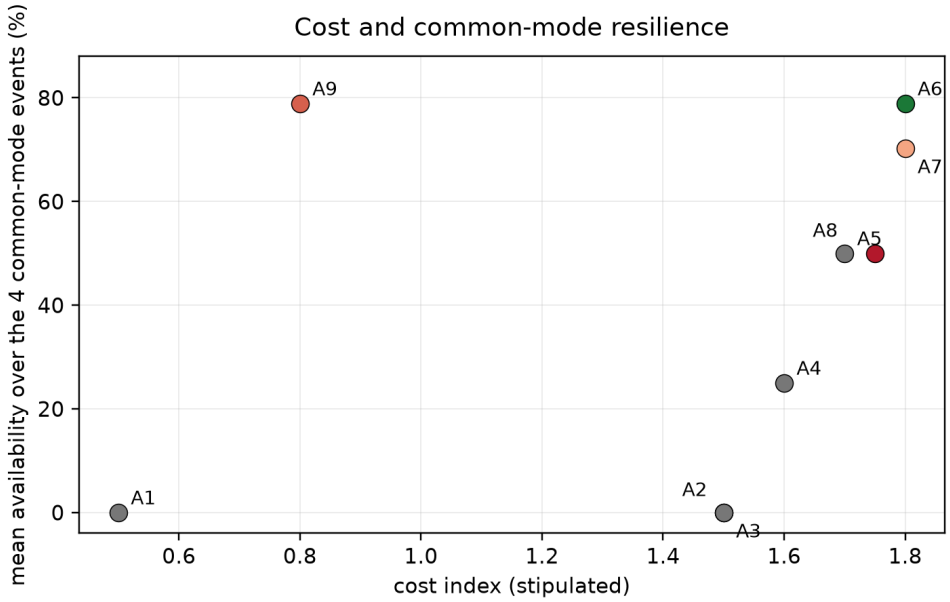


Figure 3. Cost index against mean availability over the four common-mode events (identity, AI, tenant gateway, backhaul cut). A9 matches A6 on this metric at a lower cost; A6 buys independent-link resilience (Section 5.7). The cost index is stipulated and is not a price.

5.8 Verdicts

Table 4. Pre-registered hypotheses (30-seed means).

Hypothesis and criterion	Value	Threshold	Verdict
H16 tenant isolation: A4-A3 tenant gateway; abs(A4-A3) identity	100.0; 0.0	>= 50; <= 5	PASS
H17 independence: A3-A2 SRLG; A2-A1 link	100.0; 100.0	>= 20; >= 50	PASS
H18 tested fallback: A6-A5 AI; A6 AI; A5-A6 normal	90.0; 90.0; 10.0	>= 50; >= 80; <= 15	PASS
H19 interaction: (A6-A3)-(A9-A1) AI + link	44.38	>= 20	PASS
H20 controller locality: A5-A8 controller + link; abs(A5-A8) normal	100.0; 0.0	>= 50; <= 5	PASS

Every criterion passes on the 30-seed mean and on every individual seed (pass fraction 1.000). The per-seed pass fractions and all twelve sub-criteria are in `output/common-mode/common_mode_verdicts.csv`. The all-pass result is a property of the structural model: the mechanisms are isolated by design, so the margins are large. It is not evidence that a production deployment will see the same numbers.

6. Discussion

6.1 What the comparison shows

The comparison supports three claims.

First, **current cloud isolation is the wrong cut for common-mode failure.** Tenant isolation, confidential computing, regions and provider diversity each remove a real failure mode, and the experiment confirms that they do: A4 fixes the tenant gateway, A3 fixes the conduit, A5 fixes identity. But none of them cuts the shared AI/control-plane dependency. The result is not an argument against tenant isolation; it is an argument against treating tenant isolation as sufficient for a provider-level dependency.

Second, **multipath must be independent, not merely multiple.** A2 and A3 have the same path count and opposite outcomes under the conduit failure. The path independence index is a better predictor than the number of paths. A path set that shares a control plane can even amplify a correlated failure, because every path depends on the same identity or AI endpoint.

Third, **the controller is a dependency.** A8 behaves like A5 until its controller fails, then loses the multipath advantage exactly when it is needed. A global cooperative controller is a new common-mode carrier; locality is a design variable. This is the cooperative-backhaul form of the concentration problem: the mechanism that makes self-scaling possible is also the mechanism that can centralize it.

6.2 Design implications

The results translate into five design rules.

1. **Inventory shared dependencies, then cut the control plane, not only the tenant boundary.** A tenant-isolation review that does not include identity, DNS, the AI endpoint and the orchestrator is incomplete.
2. **Measure path independence, not path count.** Report SRLG membership, provider control-plane membership and shared control-service membership for every path set. A path set with one SRLG is one path.
3. **Treat the fallback as a tested state.** Record activation probability, delay and capacity. An untested fallback is a configuration, not a mitigation; in the model it was worth 34.5 points less than a tested one.

4. **Keep cooperative control local or partition-tolerant.** A global controller can be part of the data plane's efficiency story, but it must not be on the availability path. If it is, the controller is a single point of failure with a scheduling interface.
5. **Budget the normal-operation cost of isolation.** The one-way boundary cost 10 points in this model. That is the price of local policy; it should be stated, not hidden.

6.3 The A9/A6 trade and the meaning of "both"

The cost-resilience comparison is deliberately not a single ranking. A9 (isolation only) matches A6 on the common-mode mean at less than half the cost, because both use the same boundary and fallback. A6 is better only when an independent link failure is also in scope (90% vs 45.6%). This is a useful negative result for the strongest form of the "both axes" thesis: isolation plus a tested fallback can be sufficient for a common-mode-only failure model. The argument for multipath is conditional on independent path failures being part of the threat model. The paper's claim is therefore narrower and more defensible: the two axes are independent mitigations, and the right mix depends on the failure mix, not on a universal ranking.

6.4 Relation to the motivating outages

The two outages are used as motivating examples, not as evidence. The FAA event is reported as a telecommunications capacity failure; the Microsoft event as a failed network capacity upgrade that coincided with AI service disruptions. The structural comparison does not depend on either root cause. What the testbed adds is a controlled demonstration of the mechanism: a shared dependency, a boundary that does not cut it, and a path set that does not survive it.

7. Threats to Validity

Construct validity. Availability is a capacity/dependency calculation, not a packet-level measurement. It ignores queueing, congestion control, TCP retransmission, reordering and application behaviour. The 0.9 local-policy efficiency and the 0.5/0.3 fallback capacities are stipulated. The cost index is an accounting device, not a price. The path independence index treats SRLG and provider identity as the only independence dimensions; a real deployment may have more.

Internal validity. The model is deterministic given a seed, and the 30 seeds vary event timing, demand and fallback draws, not the topology. All five hypotheses pass with large margins because the arms isolate the mechanisms by design. This is a demonstration under a frozen scenario, not a statistical estimate of an effect size. The arms differ only in mechanism and share the same draws, which rules out a seed confound; it does not rule out a scenario confound. No constant was changed after the freeze.

External validity. The scenario has six tenants, three aggregation sites, three providers and two clouds. It is not a model of a specific cloud, and the current-cloud mapping in Table 1 is a structural analogy. Different SRLG groupings, control-plane layouts or demand levels could change the levels. The qualitative ordering is the claim; the exact percentages are scenario-specific.

Conclusion validity. The hypotheses were frozen before the simulator existed, and the verdict rules are in the benchmark. A refutation would have been reported. The all-pass outcome is a limitation as well as a result: the scenario was designed to make the mechanisms observable, so it is not a random sample of architectures. The real-stack extension is the next fidelity and is not claimed here.

Security. The data diode is modelled as an availability cut, not as a cryptographic control. No key management, authentication strength or side-channel behaviour is evaluated. The one-way boundary's security properties are out of scope.

8. Conclusion

Concentration is the risk. A shared dependency with no independent failure domain behind it turns a local fault into a correlated outage, and the modern cloud stack is as capable of this as the legacy stack. Isolation

and multipath are the two independent mitigations. Isolation stops a fault at a boundary; multipath keeps service running across independent failure domains. Neither substitutes for the other, and a tested fallback is the condition that makes either one real.

The testbed supports the structural claims under its frozen scenario. Tenant isolation fixes the tenant gateway but not identity or AI; three paths over one conduit fail together; a one-way boundary with a tested fallback holds 90% under an AI failure; the isolation-multipath interaction is +44.4 points; and a global cooperative controller loses 100 points when it fails. The comparison with stylized current-cloud architectures shows the gap: current cloud improves along one axis at a time, while no current-cloud arm reaches 90% under a combined AI and link failure. The qualification is equally important: isolation plus a tested fallback can match the combined architecture on common-mode events at lower cost, so multipath is justified by independent path failures, not by common-mode failure alone.

The next step is a real-stack variant on Linux network namespaces: two disjoint paths, ECMP, link-down and `tc` outages, and an nftables control-plane cut, compared against this flow-level reference. The structural result is a map of which cuts and which paths matter; the real-stack test is how to find out what the map costs in packets.

References

1. P. Mell and T. Grance, "The NIST Definition of Cloud Computing," NIST Special Publication 800-145, 2011.
2. Amazon Web Services, "Regions and Availability Zones," AWS Documentation. [Online]. Available: AWS documentation
3. Microsoft, "Availability zones overview," Azure Reliability Documentation. [Online]. Available: Azure documentation
4. Google Cloud, "Regions and zones," Compute Engine Documentation. [Online]. Available: Google Cloud documentation
5. A. Avizienis, J.-C. Laprie, B. Randell and C. Landwehr, "Basic concepts and taxonomy of dependable and secure computing," *IEEE Transactions on Dependable and Secure Computing*, vol. 1, no. 1, pp. 11-33, 2004.
6. H. S. Gunawi, M. Hao, T. Leesatapornwongsa, T. Patana-anake, T. Do, J. Adityatama, K. J. Eliazar, A. Laksono, J. F. Lukman, V. Martin and A. D. Satria, "What Bugs Live in the Cloud? A Study of 3000+ Issues in Cloud Systems," in *Proceedings of the ACM Symposium on Cloud Computing (SoCC)*, 2014.
7. A. Basiri, N. Behnam, R. de Rooij, L. Hochstein, L. Kosewski, J. Reynolds and C. Rosenthal, "Chaos Engineering," *IEEE Software*, vol. 33, no. 3, pp. 35-41, 2016.
8. National Institute of Standards and Technology, "Guide to Operational Technology (OT) Security," NIST Special Publication 800-82 Revision 3, 2023.
9. Siemens Mobility, "Protection of OT systems," and "CoreShield Data Capture Unit." [Online]. Available: Siemens Mobility and CoreShield DCU
10. A. Ford, C. Raiciu, M. Handley, O. Bonaventure and C. Paasch, "TCP Extensions for Multipath Operation with Multiple Addresses," RFC 8684, 2020.
11. C. Hopps, "Analysis of an Equal-Cost Multi-Path Algorithm," RFC 2992, 2000.
12. K. Kompella and Y. Rekhter, "Routing Extensions in Support of Generalized Multi-Protocol Label Switching (GMPLS)," RFC 4202, 2005.
13. M. Nikolic, "Which Rewrites Survive? Pre-registered Tests of a Field-Class Model of Encapsulation, Middleboxes and Decapsulation on Linux," repository paper, 2026. [Online]. Available: Paper A
14. M. Nikolic, "Why Fair-Queueing Beats the Fick Principle," repository paper, 2026. [Online]. Available: Paper B
15. United Press International, "FAA lifts nationwide ground stop after telecommunications outage," 13 November 2026. [Online]. Available: UPI
16. Texas Public Radio, "Old tech blamed for the ground stop that caused flight chaos: 'It got overloaded'," 13 November 2026. [Online]. Available: TPR
17. DigitalToday, "Microsoft massive service outage: no evidence of malicious activity, blamed on a network upgrade error," 14 November 2026, citing TechCrunch. [Online]. Available: DigitalToday

18. Business Insider Spain, "ChatGPT y Claude caídos: el apagón masivo de Microsoft arrastra en cadena a la IA," November 2026. [Online]. Available: Business Insider
19. L. A. Barroso, J. Clidaras and U. Hölzle, *The Datacenter as a Computer: Designing Warehouse-Scale Machines*, 3rd ed. Morgan and Claypool, 2018.
20. National Institute of Standards and Technology, "Hardware-Enabled Security for Server Platforms: Security Architecture and Design," NIST IR 8320, 2021.

Appendix: Artifact Index

Artifact	Path
Frozen protocol	`docs/common-mode-predictions.md`
Results and deviations	`docs/common-mode-results.md`
Testbed guide	`docs/common-mode-testbed.md`
Simulator	`simulation/common_mode.py`
Benchmark	`tools/benchmark_common_mode.py`
Figures	`tools/make_common_mode_figures.py`
Tests	`tests/test_common_mode.py`